DO NOW — RECALL

Lesson 094 · CPU Performance: Clock, Cores and Cache

Name: _____ Class: _____ Date: _____

1. What is a cache hit?

2. What is a cycle?

3. What is a core?

4. A search space of 64 candidates is repeatedly halved to one. How many halvings?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Requested item found in cache.

2. Model answer

A unit of processor clock timing.

3. Model answer

An independent CPU execution unit.

4. Model answer

64→32→16→8→4→2→1; six halvings.

By the end, you can…

• Clock speed counts clock cycles per second; one cycle is not always one complete instruction. Frequency alone cannot predict program performance.

• Multiple CPU cores improve performance most for work divided into independently executable threads; a strictly sequential task may gain little.

• Cache is small, fast storage close to CPU. A cache hit avoids some slower RAM accesses; cache size, locality and hit rate influence results.

Vocabulary — know these terms

GHz — Billions of cycles per second.

core — CPU execution core.

cache hit — Data found in CPU cache.

IPC — Instructions retired per cycle (workload dependent).

NEW LEARNING

Read, annotate and explain

Observe

Clock speed counts clock cycles per second; one cycle is not always one complete instruction. Frequency alone cannot predict program performance.

Explain

Multiple CPU cores improve performance most for work divided into independently executable threads; a strictly sequential task may gain little.

Evaluate

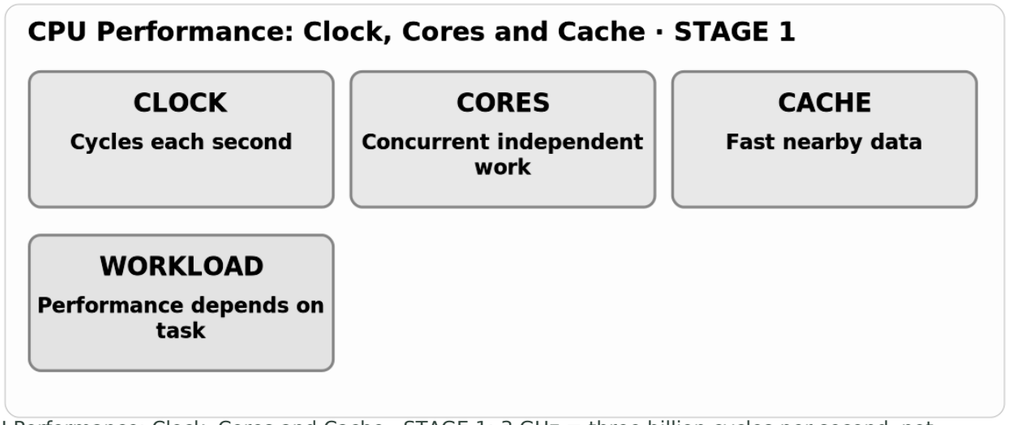
Cache is small, fast storage close to CPU. A cache hit avoids some slower RAM accesses; cache size, locality and hit rate influence results.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

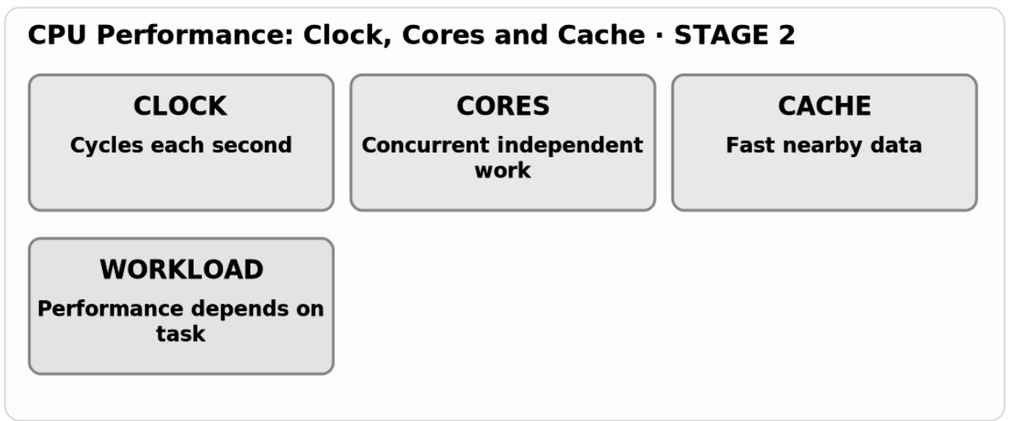
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



CPU Performance: Clock, Cores and Cache · STAGE 1; 3 GHz = three billion cycles per second, not instructions

New Learning



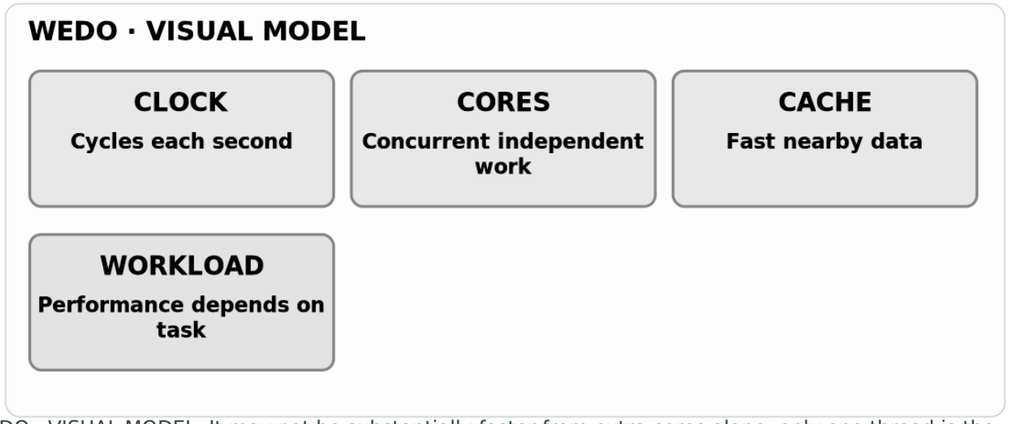
CPU Performance: Clock, Cores and Cache · STAGE 2; 3 GHz = three billion cycles per second, not instructions; Independent cores can process suitable threads

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A single-threaded algorithm runs on a 2-core or 8-core CPU of equal per-core performance. Explain likely benef

It may not be substantially faster from extra cores alone; only one thread is the bottleneck.



WEDO · VISUAL MODEL; It may not be substantially faster from extra cores alone; only one thread is the bottleneck.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

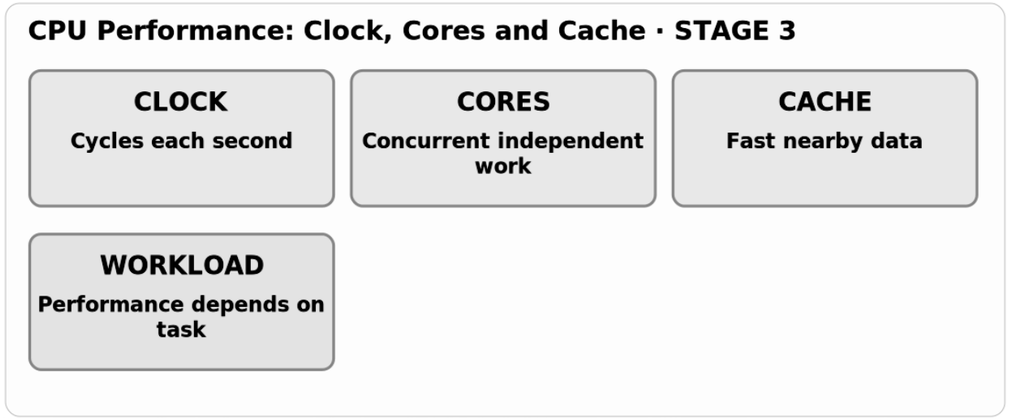
Guided 1 — CHECK

Explain the mechanism
It may not be substantially faster from extra cores alone; only one thread is the bottleneck.
Parallel benefits need tasks/threads that can run concurrently.

New Learning · original lesson diagrams

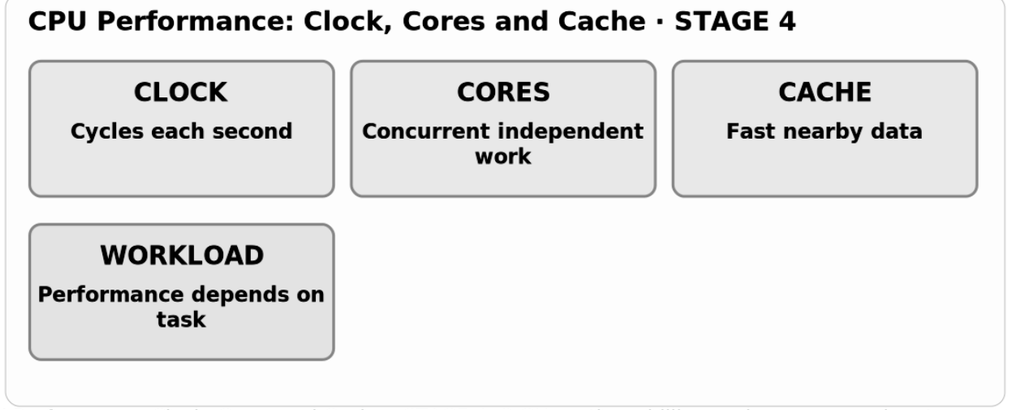
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



CPU Performance: Clock, Cores and Cache · STAGE 3; 3 GHz = three billion cycles per second, not instructions; Independent cores can process suitable threads; Fast small memory may avoid slower main RAM reads

New Learning



CPU Performance: Clock, Cores and Cache · STAGE 4; 3 GHz = three billion cycles per second, not instructions; Independent cores can process suitable threads; Fast small memory may avoid slower main RAM reads; Workload, IPC, contention and cache misses affect throughput

I DO — WORKED EXAMPLE

Follow the worked method

Compare 2 GHz and 3 GHz clock rates under equal work per clock and no bottlenecks.

Explain the mechanism

3 GHz offers 50% more cycles each second than2 GHz; this does not universally imply 50% faster program runtime.

The comparison requires equal microarchitecture and workload assumptions.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A 4GHz processor always completes any program twice as fast as a 2GHz processor.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

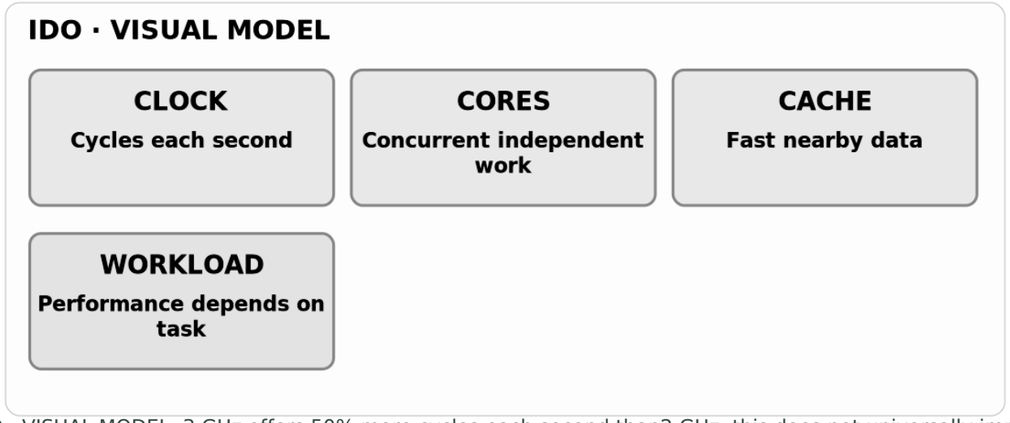
Recommend improvements for school video rendering versus a sequential exam-marking script. Discuss CPU performance factors. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Compare 2 GHz and 3 GHz clock rates under equal work per clock and no bottlenecks.

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IDO · VISUAL MODEL; 3 GHz offers 50% more cycles each second than2 GHz; this does not universally imply 50% faster program runtime.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A single-threaded algorithm runs on a 2-core or 8-core CPU of equal per-core performance. Explain likely benefit.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Explain effects of clock speed, number of cores and cache on performance. [6]

Question 2

CPU X 2GHz and Y 3GHz have identical IPC; compute ratio of cycles/sec and one reason runtime may differ from ratio. [6]